

14 A car is driven over a series of equally spaced bumps separated by a distance of 5.0 m.

If the natural frequency of the car's suspension system is 2.0 Hz, what is the speed of the car for it to experience a large amplitude of vibration?

A 1.0 m s^{-1}

B 2.5 m s^{-1}

C 5.0 m s^{-1}

D 10 m s^{-1}

15 A tuning fork is made to vibrate above a tube filled with water. The water is allowed to run